

A Practical Guide to Feature selection

TOPIC -- FEATURE SELECTION

-- 01 --

Hilbert-Schmidt Independence Criterion Lasso based feature selection For high-dimensional and small sample data (e.g., dimensionality > 105 and the number of samples < 103), the Hilbert-Schmidt Independence Criterion Lasso (HSIC Lasso) is useful. HSIC Lasso optimization problem is given as

-- 02 --

$$\text{QPFS} : \min_{\mathbf{x}} \{ \alpha \mathbf{x}^T \mathbf{H} \mathbf{x} - \mathbf{x}^T \mathbf{F} \} \text{ s.t. } \sum_{i=1}^n x_i = 1, x_i \geq 0$$
$$\text{QPFS} : \min_{\mathbf{x}} \{ \alpha \mathbf{x}^T \mathbf{H} \mathbf{x} - \mathbf{x}^T \mathbf{F} \} \text{ s.t. } \sum_{i=1}^n x_i = 1, x_i \geq 0$$

-- 03 --

The score uses the conditional mutual information and the mutual information to estimate the redundancy between the already selected features ($f_j \in S$) and the feature under investigation (f_i).

-- 04 --

The mRMR algorithm is an approximation of the theoretically optimal maximum-dependency feature selection algorithm that maximizes the mutual information between the joint distribution of the selected features and the classification variable. As mRMR approximates the combinatorial estimation problem with a series of much smaller problems, each of which only involves two variables, it thus uses pairwise joint probabilities which are more robust. In certain situations the algorithm may underestimate the usefulness of features as it has no way to measure interactions between features which can increase relevancy. This can lead to poor performance when the features are individually useless, but are useful when combined (a pathological case is found when the class is a parity function of the features). Overall the algorithm is more efficient (in terms of the amount of data required) than the theoretically optimal max-dependency selection, yet produces a feature set with little pairwise redundancy. mRMR is an instance of a large class of filter methods which trade off between relevancy and redundancy in different ways.

-- 05 --

Joint mutual information In a study of different scores Brown et al. recommended the joint mutual information as a good score for feature selection. The score tries to find the feature, that adds the most new information to the already selected features, in order to avoid redundancy. The score is formulated as follows:

-- 06 --

Further reading Guyon, Isabelle; Elisseeff, Andre (2003). "An Introduction to Variable and Feature Selection". *Journal of Machine Learning Research*. 3: 1157–1182. Harrell, F. (2001). *Regression Modeling Strategies*. Springer. ISBN 0-387-95232-2. Liu, Huan; Motoda, Hiroshi (1998). *Feature Selection for Knowledge Discovery and Data Mining*. Springer. ISBN 0-7923-8198-X. Liu, Huan; Yu, Lei (2005). "Toward Integrating Feature Selection Algorithms for Classification and Clustering". *IEEE Transactions on Knowledge and Data Engineering*. 17 (4): 491–502. Bibcode:2005IDSO...17..491L. doi:10.1109/TKDE.2005.66. S2CID 1607600.

-- 07 --

Optimality criteria The choice of optimality criteria is difficult as there are multiple objectives in a feature selection task. Many common criteria incorporate a measure of accuracy, penalised by the number of features selected. Examples include Akaike information criterion (AIC) and Mallows's Cp, which have a penalty of 2 for each added feature. AIC is based on information theory, and is effectively derived via the maximum entropy principle. Other criteria are Bayesian information criterion (BIC), which uses a penalty of $\log \sqrt{n}$ for each added feature, minimum description length (MDL) which asymptotically uses $\log \sqrt{n}$, Bonferroni / RIC which use $2 \log \sqrt{p}$, maximum dependency feature selection, and a variety of new criteria that are motivated by false discovery rate (FDR), which use something close to $2 \log \sqrt{\frac{p}{q}}$. A maximum entropy rate criterion may also be used to select the most relevant subset of features.

-- 08 --

simplification of models to make them easier to interpret, shorter training times, to avoid the curse of dimensionality, improve the compatibility of the data with a certain learning model class, to encode inherent symmetries present in the input space. The central premise when using feature selection is that data sometimes contains features that are redundant or irrelevant, and can thus be removed without incurring much loss of information. Redundancy and irrelevance are two distinct notions, since one relevant feature may be redundant in the presence of

another relevant feature with which it is strongly correlated. Feature extraction creates new features from functions of the original features, whereas feature selection finds a subset of the features. Feature selection techniques are often used in domains where there are many features and comparatively few samples (data points).

-- 09 --

$$R(S) = \frac{1}{|S|^2} \sum_{f_i, f_j \in S} I(f_i; f_j) \quad \{\displaystyle R(S) = \frac{1}{|S|^2} \sum_{f_i, f_j \in S} I(f_i; f_j)\}$$

-- 10 --

Regularized trees The features from a decision tree or a tree ensemble are shown to be redundant. A recent method called regularized tree can be used for feature subset selection. Regularized trees penalize using a variable similar to the variables selected at previous tree nodes for splitting the current node. Regularized trees only need build one tree model (or one tree ensemble model) and thus are computationally efficient. Regularized trees naturally handle numerical and categorical features, interactions and nonlinearities. They are invariant to attribute scales (units) and insensitive to outliers, and thus, require little data preprocessing such as normalization. Regularized random forest (RRF) is one type of regularized trees. The guided RRF is an enhanced RRF which is guided by the importance scores from an ordinary random forest.

-- 11 --

where $F_{n \times 1} = [I(f_1; c), \dots, I(f_n; c)]^T$ is the vector of feature relevancy assuming there are n features in total, $H_{n \times n} = [I(f_i; f_j)]_{i,j=1 \dots n}$ is the matrix of feature pairwise redundancy, and $\mathbf{x}_{n \times 1}$ represents relative feature weights. QPFS is solved via quadratic programming. It is recently shown that QPFS is biased towards features with smaller entropy, due to its placement of the feature self redundancy term $I(f_i; f_i)$ on the diagonal of H .